Md Tameem Rahman

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Education

Rajshahi University of Engineering & Technology, Rajshahi *March 2022 – August 2026*
BSc in Computer Science & Engineering

New Govt. Degree College, Rajshahi *2018 – 2020*
Higher Secondary Certificate (HSC), Science

Govt. Laboratory High School, Rajshahi *2010 – 2018*
Secondary School Certificate (SSC), Science

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Structures & Algorithms, Object-Oriented Programming, Computer Networks, Security & Privacy, Database Systems, Software Engineering

Technical Skills

Programming Languages: C, C++, Python, JavaScript

ML / DL Frameworks & Libraries: Scikit-learn, OpenCV, PyTorch, TensorFlow, Optuna

LLM / AI: LangChain, LangGraph, LangSmith, Ollama, RAG, Model Context Protocol (MCP)

ML Deployment & APIs: FastAPI, Streamlit

Web Development: HTML, CSS, React, Next.js, Node.js, Express.js, Tailwind CSS

Databases: MySQL, PostgreSQL, SQLite, MongoDB

Tools & Technologies: Git, GitHub, Linux, Firebase

Experience

Industrial Attachment Trainee, Chorcha EdTech Ltd., Rajshahi *March 2025 – April 2025*

- Developed LLM-based MCQ validation and doubt-solving systems, and a scalable geolocation solution for identifying Bangladesh's administrative divisions from coordinates; gained experience in prompt engineering, LLM application development, and structured data processing.

Technologies: Python, Large Language Models (LLMs), Prompt Engineering, Geospatial Data Processing

Leadership & Extracurricular

Joint Secretary, Rajshahi City Association, RUET *July 2025 – July 2026*

Organizing Secretary, RUET Computing Society *December 2025 – August 2026*

Treasurer, Lab'18 Alumni Association, Rajshahi Govt. Laboratory High School, Rajshahi *June 2022 – Present*

Research & Thesis

Privacy-Preserving Federated Intrusion Detection Across Heterogeneous Networks *October 2025 – August 2026*
Thesis, Dept. of CSE, RUET

- Designing CNN-based NIDS models trained across distributed clients without sharing raw network traffic data. Addressing concept shift via selective parameter sharing to improve detection performance while preserving data privacy.
- Achieved 93.23% accuracy and 0.9689 AUC-ROC on the final model

Image Forgery Detection *July 2025 – March 2026*

- Designed a dual-branch CNN combining a pretrained ResNet18 RGB stream with a noise-residual branch (high-pass filtering + custom CNN encoder) to capture both semantic and low-level forensic artifacts.
- Trained on the Defacto dataset with class-imbalance-aware loss weighting, achieving 98.9% accuracy and 0.999 AUC-ROC on in-domain evaluation.
- Evaluated cross-dataset generalization across four standard benchmarks (CASIAv1, Columbia, Coverage, IMD2020), building a full evaluation pipeline (confusion matrices, ROC curves) to analyze domain-shift behavior.

Academic Projects

Diabetes Prediction App *January 2025 – February 2025*

Streamlit, Scikit-learn, Joblib, TensorFlow, SQLite

- Developed an end-to-end machine learning–based application to predict diabetes risk using patient health data.
- Designed a prediction pipeline to process new user inputs and generate real-time inference from trained models.
- Integrated a secure user authentication system using SQLite and password hashing.

Car Doctor Pro — Full-Stack Auto Servicing Booking Platform *October 2024 – January 2025*

Next.js 14(App Router), React, MongoDB Atlas, NextAuth.js, Tailwind CSS, DaisyUI

- Engineered a full-stack automotive service booking web application with Next.js 14 App Router and MongoDB Atlas, featuring live search filtering and dynamic service catalogs.
- Implemented secure multi-provider authentication (OAuth via Google/GitHub credentials with bcryptjs) and protected private routes using Next.js Edge Middleware.
- Designed end-to-end booking lifecycle management with full CRUD operations, allowing authenticated users to schedule, update, and manage service appointments in real time.